

Geometry — revision of Test 1

Lessons 1–6 · Quarter I · 40 questions

Sketch every question before answering it, and name the fact you used — the angle sum, the property, or the test. A number with no reason earns no method marks.

Lessons 1–2 · Revision of the Grade 7 course 9

Angles and triangles

- 1

Two angles of a triangle are 42° and 63° . Find the third.
 75°
The angle sum of a triangle is 180° .
- 2

An isosceles triangle has a base angle of 55° . Find the apex angle.
 70°
 $180^\circ - 2 \times 55^\circ$.
- 3

An isosceles triangle has an apex angle of 96° . Find each base angle.
 42°
The two base angles are equal: $(180^\circ - 96^\circ) \div 2$.
- 4

Two interior angles of a triangle are 40° and 65° . Find the exterior angle at the third vertex.
 105°
An exterior angle equals the sum of the two remote interior angles.
- 5

One acute angle of a right-angled triangle is 37° . Find the other.
 53°
The two acute angles add to 90° .
- 6

State each angle of an equilateral triangle.
 60°
- 7

Two angles lie on a straight line and one is 118° . Find the other.
 62°
Angles on a straight line add to 180° .
- 8

Two straight lines cross. One angle is 74° . Find the angle vertically opposite it.
 74°
Vertically opposite angles are equal.
- 9

The angles of a triangle are in the ratio $2 : 3 : 4$. Find them.
 $40^\circ, 60^\circ, 80^\circ$
Nine parts make 180° , so one part is 20° .

Lessons 3–4 · Polygons. Interior and exterior angles 14

Angle sums, regular polygons

- 10

Find the sum of the interior angles of a hexagon.
 720°
 $(6 - 2) \times 180^\circ$.
- 11

Find the sum of the interior angles of a decagon.
 1440°
 $(10 - 2) \times 180^\circ$.
- 12

The interior angles of a convex polygon add to 900° . How many sides?
 7
 $(n - 2) \times 180^\circ = 900^\circ$.
- 13

The interior angles of a convex polygon add to 2160° . How many sides?
 14
- 14

Each exterior angle of a regular polygon is 24° . How many sides?
 15
The exterior angles add to 360° .
- 15

Each exterior angle of a regular polygon is 72° . How many sides?
 5
- 16

Find each interior angle of a regular pentagon.
 108°
 $540^\circ \div 5$, or $180^\circ - 72^\circ$.
- 17

Find each interior angle of a regular octagon.
 135°
- 18

Each interior angle of a regular polygon is 150° . How many sides?
 12
Then each exterior angle is 30° , and $360^\circ \div 30^\circ = 12$.
- 19

What do the exterior angles of *any* convex polygon add to?
 360°
The number of sides does not matter.
- 20

A regular polygon has 20 sides. Find each exterior angle.
 18°
- 21

Four angles of a pentagon are $100^\circ, 110^\circ, 120^\circ$ and 95° . Find the fifth.
 115°
The five add to 540° .
- 22

The interior angles of a convex polygon add to 1260° . How many sides?
 9
- 23

Can a regular polygon have an exterior angle of 50° ? Explain.
no
 $360^\circ \div 50^\circ = 7.2$, and the number of sides must be a whole number.

Lesson 5 · Parallelogram and its properties 9

Opposite sides and angles, diagonals

- 24

In parallelogram $ABCD$, $\angle A = 68^\circ$. Find $\angle B$, $\angle C$ and $\angle D$.
 $\angle B = 112^\circ, \angle C = 68^\circ, \angle D = 112^\circ$
Opposite angles are equal; neighbouring angles are supplementary.
- 25

The perimeter of a parallelogram is 42 cm and one side is 12 cm. Find the other side.
 9 cm
Two sides of each length: $42 \div 2 - 12$.
- 26

A parallelogram has $AB = 7\text{ cm}$ and $BC = 11\text{ cm}$. Find its perimeter.
 36 cm
- 27

The diagonal AC of parallelogram $ABCD$ is 16 cm and the diagonals meet at O. Find AO.
 8 cm
The diagonals bisect each other.
- 28

In parallelogram $ABCD$, $\angle A : \angle B = 2 : 3$. Find both angles.
 $\angle A = 72^\circ, \angle B = 108^\circ$
They are supplementary, so 5 parts make 180° .
- 29

One angle of a parallelogram is 90° . What can you say about the other three?
all 90° — it is a rectangle
Its neighbour is $180^\circ - 90^\circ$, and opposite angles repeat.

- 30 In parallelogram $ABCD$ the diagonal BD makes $\angle ABD = 35^\circ$ and $\angle ADB = 60^\circ$. Find $\angle A$.

85°

The three angles of $\triangle ABD$ add to 180° .

- 31 What do two neighbouring angles of a parallelogram add to?

180°

They are co-interior angles between the parallel sides.

- 32 In parallelogram $ABCD$, $\angle A = 3\angle B$. Find both angles.

$\angle A = 135^\circ$, $\angle B = 45^\circ$

They are supplementary: $3B + B = 180^\circ$.

Lesson 6 · Tests for a parallelogram 8

Which test proves it — and which does not

- 33 In $ABCD$, $AB \parallel CD$ and $AB = CD$. Is it a parallelogram? Name the test.

yes — one pair of sides equal and parallel

- 34 In $ABCD$, $AB = CD$ and $BC = AD$. Is it a parallelogram? Name the test.

yes — both pairs of opposite sides equal

- 35 In $ABCD$ the diagonals bisect each other. Is it a parallelogram?

yes — the diagonal test

- 36 In $ABCD$, $AB \parallel CD$ and $BC \parallel AD$. Is it a parallelogram?

yes — this is the definition

- 37 In $ABCD$, $AB \parallel CD$ and $BC = AD$. Must it be a parallelogram?

no

An isosceles trapezium satisfies this and is not a parallelogram — the equal pair must be the *parallel* pair.

- 38 In $ABCD$, $\angle A = \angle C$ and $\angle B = \angle D$. Is it a parallelogram?

yes — both pairs of opposite angles equal

- 39 The diagonals of $ABCD$ meet at O with $AO = OC = 5$ cm and $BO = OD = 7$ cm. Is it a parallelogram? Name the test.

yes — each diagonal is bisected by the other

- 40 In $ABCD$, $\angle A = \angle B$ and $\angle C = \angle D$. Must it be a parallelogram?

no

Those are *neighbouring* pairs. An isosceles trapezium has equal base angles and is not a parallelogram — the test needs the *opposite* pairs equal.